



Exploring relationships between map patterns of cylindrically folded layers and their stereonet

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Students should have already been briefly introduced to the anatomy of a stereonet and how planar and linear features are represented. They should already know basic features of folds: wavelength, amplitude, fold hinges and limbs, cylindrical and noncylindrical folds, etc. Students should be able to recognize and interpret strike/dip map symbology

Introduction (with geoscience context)

Geologic maps and stereographic projections are both useful tools for communicating structural data, although each has limitations regarding what elements of a geologic structure can be effectively represented. In this homework assignment, you will explore the relationship between geologic map patterns and equal-area lower hemisphere stereographic projections derived from bedding measurements contained within the same map. Successful completion of this homework will enable students to:

Learning Goals

- Strengthen visualization skills needed to investigate sections through geologic structures.
- Determine how elements of fold geometry (namely, fold axis orientation) are represented on a stereonet;
- Recognize and describe the aspects of fold geometry that cannot be communicated using a stereographic projection

Using the Visible Geology Tools

This homework assignment will utilize both the Stereonet and Geology Explorer tools. Students are encouraged to play around with both tools independently prior to starting the homework activity, but it is not required.

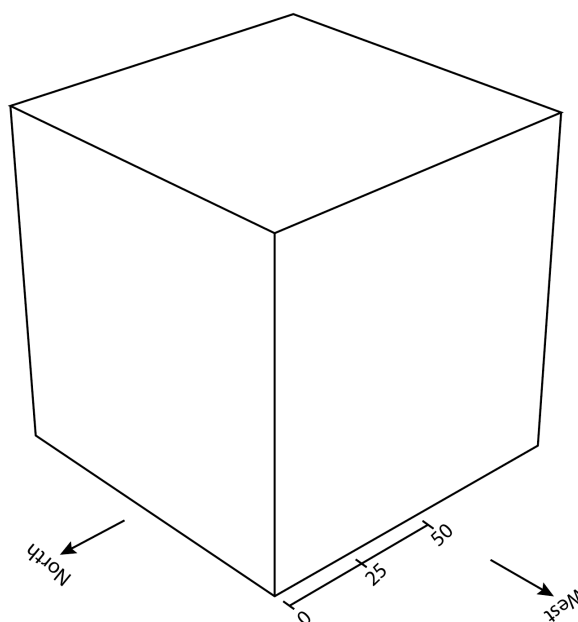
Activity (description and steps)

1. *Model 1 – Open the Geology Explorer model here:*
<https://www.visiblegeology.com?sharecode=Y7AFxxTPw4C5zufeR9bn5D>
 - a. First, get acquainted with the model: You can drag and rotate the model. Right-clicking and moving the cursor allows you to pan, and you can zoom with the scroll wheel on your mouse. Make note of the controls in the upper left that allow you to view one face of the block at a time.
 - b. Next, get acquainted with the model controls. For our purposes, you will want to know about:

- o Layers: shows you the names, thicknesses, and colors of each layer in the models. No need to alter this.
- o Events: shows you the selection of geologic events that you can apply to your model. Relevant events for this activity are tilting and folding. No need to alter these.
- o History: shows you the geologic events that have already shaped your model. Note that you can reorder events by dragging them in the history pane. You can step through the history by clicking on each event. The details of events can be edited by clicking on the pencil button of each event.

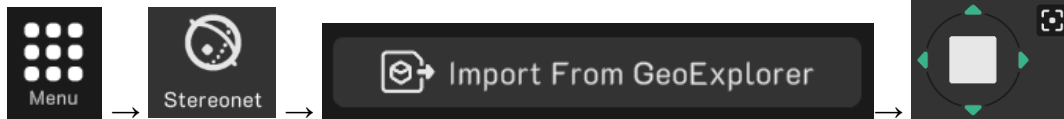
c. Reproduce the default view of the block in the form at right. Include representative strike and dip symbols.

d. Describe the map pattern created by the sedimentary layers. How do strike and dip symbols relate to the contacts between layers?



e. Identify the structure(s) and describe their geometry. Be as specific as possible.

- f. Open the stereonet program and import data from geo explorer. Look at a top-down view of the stereonet (2-D view).

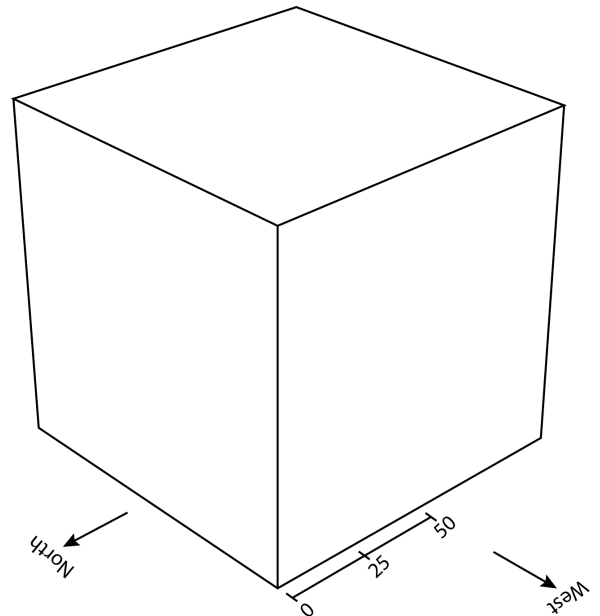


Describe the pattern of planes and poles to planes.

2. *Model 2 – Open the Geology Explorer model here:*

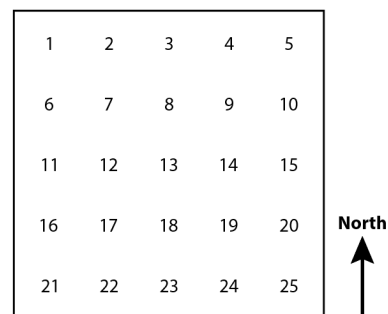
<https://www.visiblegeology.com?sharecode=QcnjHdG9FfVYpofYeQE8WX>

- a. Reproduce the default view of the block in the form at right. Include representative strike and dip symbols.
- b. Describe the map pattern created by the sedimentary layers. How do strike and dip symbols relate to the contacts between layers?



- c. Identify the structure(s) and describe their geometry. Be as specific as possible.

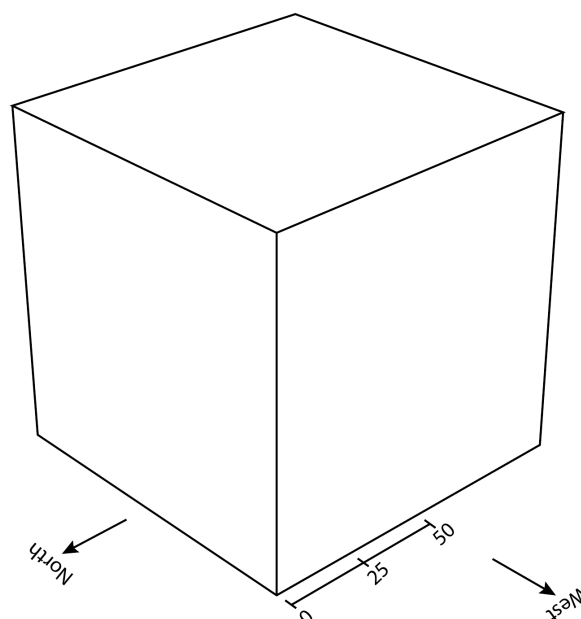
- d. Open the stereonet program and import data from geo explorer. Look at a top-down view of the stereonet (2-D view). **Describe the pattern of planes and poles to planes. Where do measurements from the southwest half of the map area plot? The northeast?** You can try to determine this visually without editing the model, but you are also free to hide some of the strike/dip symbols from view by toggling off the eye icon. The measurements are entered sequentially from 1–25 following the diagram at right.



3. *Model 3 – Open the Geology Explorer model here:*

<https://www.visiblegeology.com?sharecode=keUxKv3VsuZLx5baxhxdQQ>

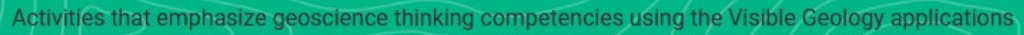
- a. Reproduce the default view of the block in the form at right. Include representative strike and dip symbols.
- b. Describe the map pattern created by the sedimentary layers. How do strike and dip symbols relate to the contacts between layers?





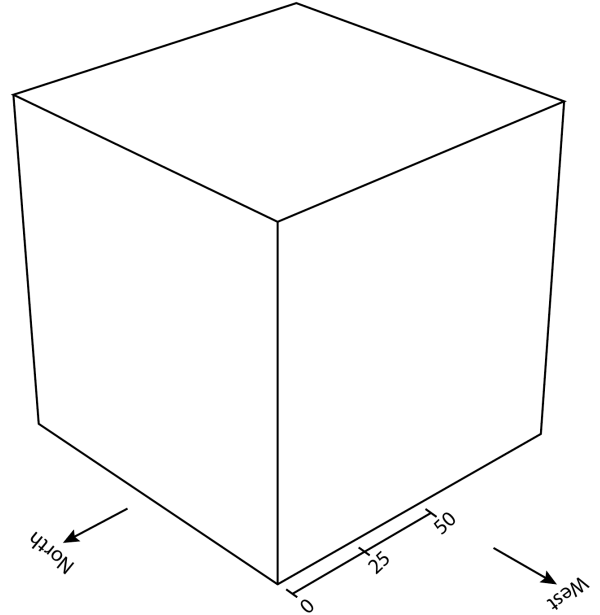
c. Identify the structure(s) and describe their geometry. Be as specific as possible.

d. Open the stereonet program and import data from geo explorer. Look at a top-down view of the stereonet (2-D view). **Describe the pattern of planes and poles to planes. Where do measurements from the southwest half of the map area plot? The northeast? Compare this pattern to the pattern for Model 2.**



<https://www.visiblegeology.com?sharecode=gG3KmVwUYwB7euQ8aeUmx2>

- b. Describe the map pattern created by the sedimentary layers. How do strike and dip symbols relate to the contacts between layers?



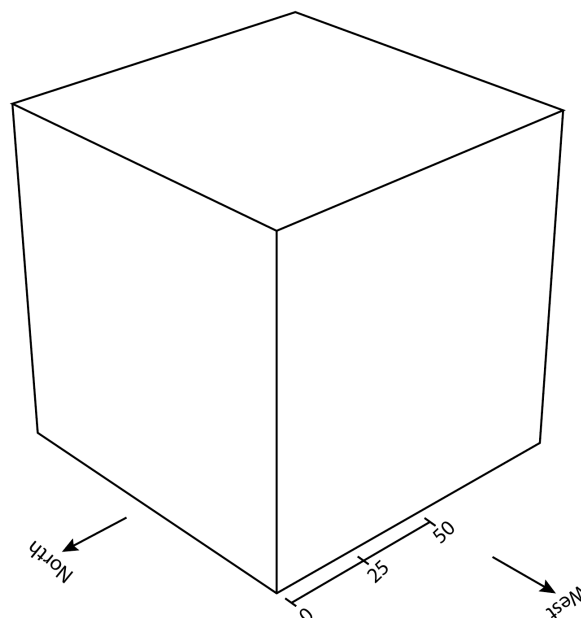
- c. Identify the structure(s) and describe their geometry. Be as specific as possible.
- d. Open the stereonet program and import data from geo explorer. Look at a top-down view of the stereonet (2-D view). **Describe the pattern of planes and poles to planes.**

Does the pattern of poles to planes have any geometric relationship to the pattern of planes? Describe. Where do measurements from the southwest half of the map area plot? The northeast? Compare this pattern to the pattern for model 3.

5. *Model 5 – Open the Geology Explorer model here:*

<https://www.visiblegeology.com?sharecode=R5wgeHpFRbC5aFRYgHdpQK>

- a. Reproduce the default view of the block in the form at right. Include representative strike and dip symbols.
- b. Describe the map pattern created by the sedimentary layers. How do strike and dip symbols relate to the contacts between layers?



- c. Identify the structure(s) and describe their geometry. Be as specific as possible.



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- d. Open the stereonet program and import data from geo explorer. Look at a top-down view of the stereonet (2-D view). Describe the pattern of planes and poles to planes. Does the pattern of poles to planes have any geometric relationship to the pattern of planes? Describe. Where do measurements from the southwest half of the map area plot? The northeast? Compare this pattern to the pattern for model 4.

6. Consider the sum of the models you have inspected

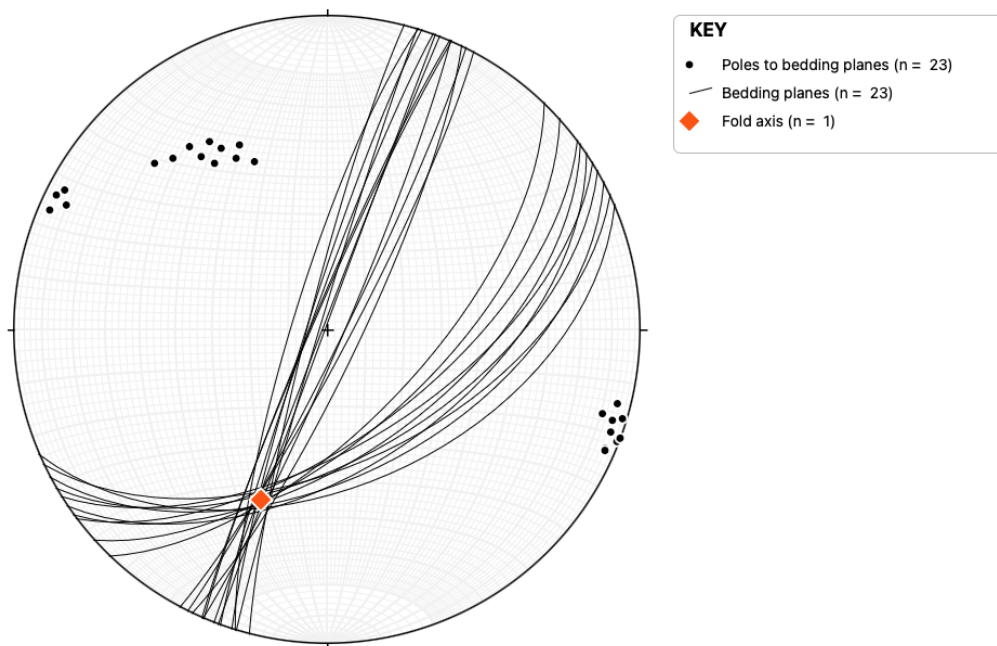
- a. What are two ways that fold axes are represented on a stereonet? Feel free to add sketches to support your description.
- b. What information present in a geologic map is lost when the same data are plotted on a stereonet?



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- c. Given your answer to question b above, what fold configurations could be represented by the stereonet below? You should be able to describe at least two. Supplement your descriptions with simplified cross section sketches below.





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